Individuals Post-Stroke Do Not Self-Optimize to Novel AFO Use After One Month

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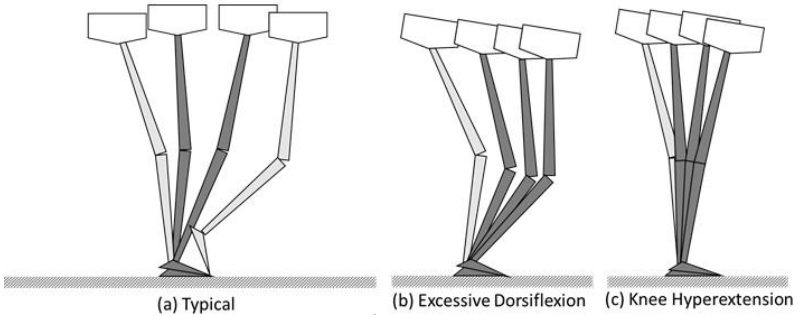
Background

- 795,000 strokes in the United States every year[1]
  - Commonly leads to plantar flexor weakness
- Plantar flexors act eccentrically to control tibial rotation during stance and to store and return energy
- Plantar flexor weakness causes walking difficulties



https://www.sportsinjuryclinic.net/anatomy/plantar-flexion

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1. cdc.gov/stroke/index.htm

PD-AFOs

- Passive-dynamic ankle-foot orthoses (PD-AFOs) have been shown to assist weakened plantar flexors[2]
- Spring-like mechanism
  - Store and return energy
- PD-AFOs used in this project were customized for each participant
  - Stiffness based on their level of plantar flexor weakness

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2. Koller, C. et al. Pros Orth Intl, 45, 313-321, 2021.(a)

Prior Work

- Stiffness-customized PD-AFOs reduce mechanical cost of transport (COT) compared to walking with either No AFO or standard of care (SOC) AFOs[3,4]

- **Are the initial benefits of PD-AFO use improved through self-acclimation?**

- Self-acclimation – allowing participants to become more familiar with the device without instruction from researchers
  - No established training method with new AFO
  - Neuromotive impairments make training difficult

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3. Arch, E.S. et al. Journal of Prosthetics and Orthotics, 28(2), 60, 2016
4. Koller, C. et al. Proc. ASB Annual Meeting, 2021.(b)

Purpose:

Investigate if individuals post-stroke self-optimized walking performance with a newly prescribed PD-AFO during one month in the free-living environment

Hypotheses:

Over the one-month period, participants would:

- ⬆ Self-selected walking speed
- ⬆ Peak plantar flexion (PF) moment during stance
- ⬆ Peak ankle dorsiflexion (DF) during stance
- ⬇ Total COT

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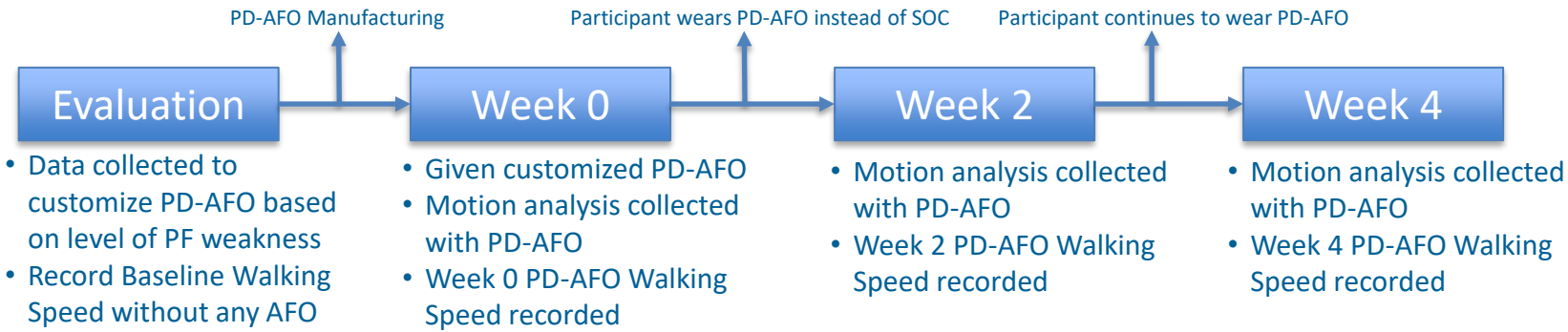
Methods

- 9 individuals (4 male, 5 female) with single unilateral chronic stroke
  - Age 64.8 (6.5) years; Height 1.73 (0.12) m; Mass 85.2 (16.3) kg



- Motion analysis performed with instrumented treadmill and motion capture cameras
- All treadmill walking done at Baseline Walking Speed

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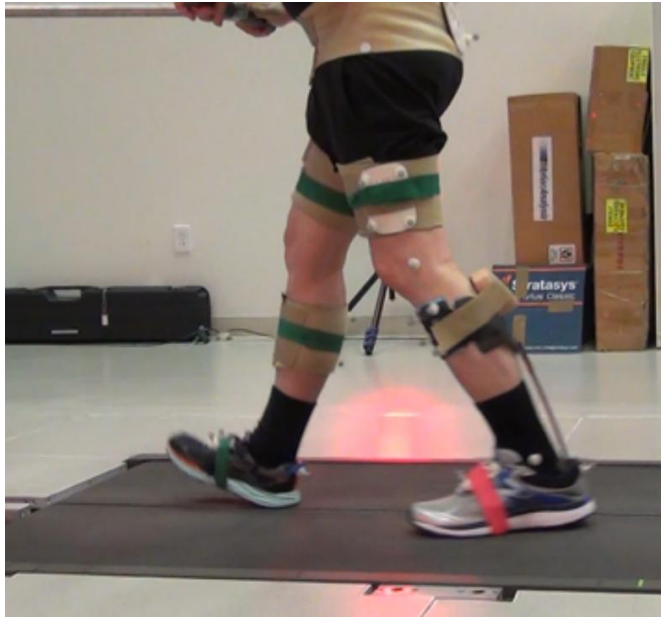
Outcome Variables

- Self-selected walking speed
  - Overground after treadmill walking
- Peak Ankle Dorsiflexion
  - Engagement of spring-like PD-AFO
- Peak Plantar Flexor Moment
  - Control of forward rotation of tibia
- Total mechanical COT

$$COT = \frac{\frac{hip^{+} + knee^{+} + ankle^{+} + foot^{+}}{mass} + abs\left(\frac{hip^{-} + knee^{-} + ankle^{-} + foot^{-}}{mass}\right)}{stride\ length}$$



Results – Synopsis

	Self-Selected Walking Speed (m/s)	Peak Ankle DF (deg)	Peak PF Moment (N*m)	Total COT (J/kg/m)
Week 0	0.76 (0.28)	17.44 (5.98)	-0.89 (0.25)	2.34 (0.40)
Week 2	0.81 (0.25)	15.11 (6.82)	-0.95 (0.31)	2.33 (0.42)
Week 4	0.79 (0.27)	16.22 (5.66)	-0.95 (0.24)	2.29 (0.42)
p	0.063	0.309	0.0997	0.354
η_p^2	0.292	0.137	0.250	0.121

X No significant differences in any outcome measures over 1 month

Participants do NOT change walking mechanics over one month of use of a novel AFO

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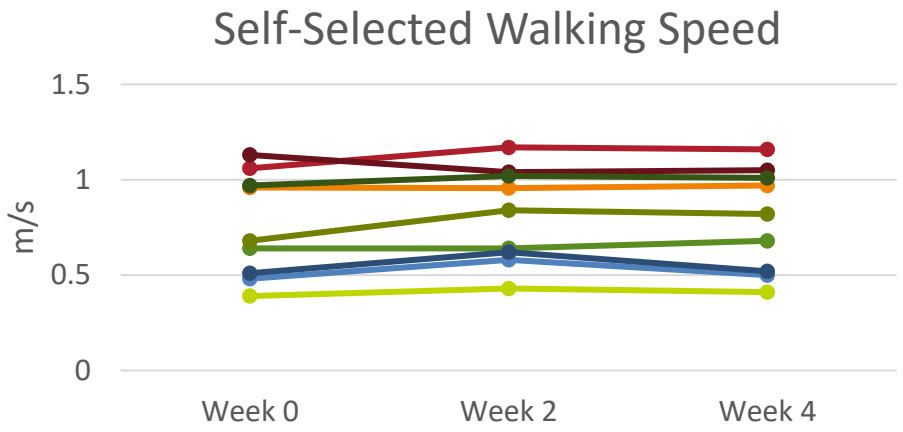
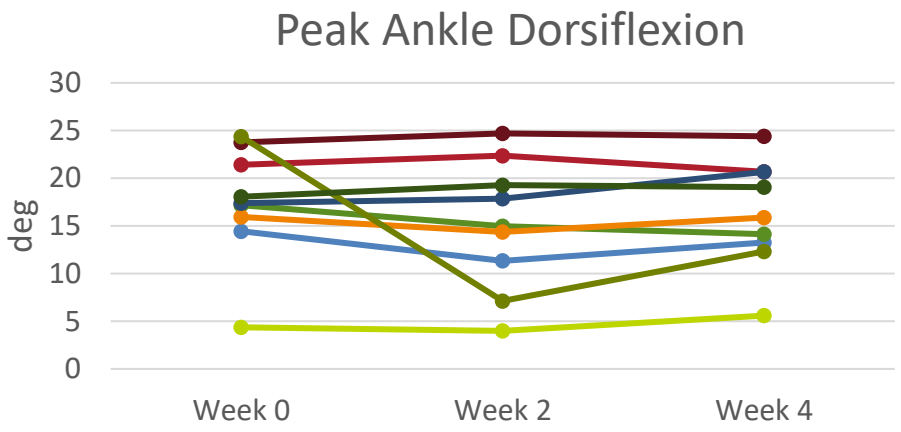
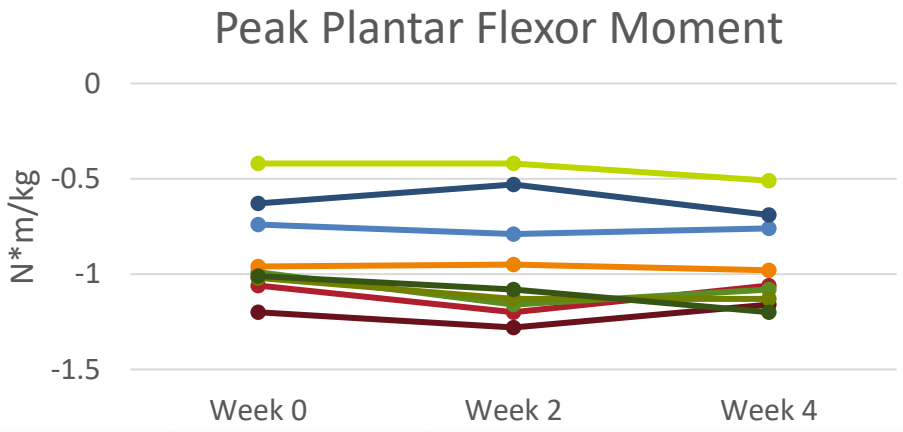
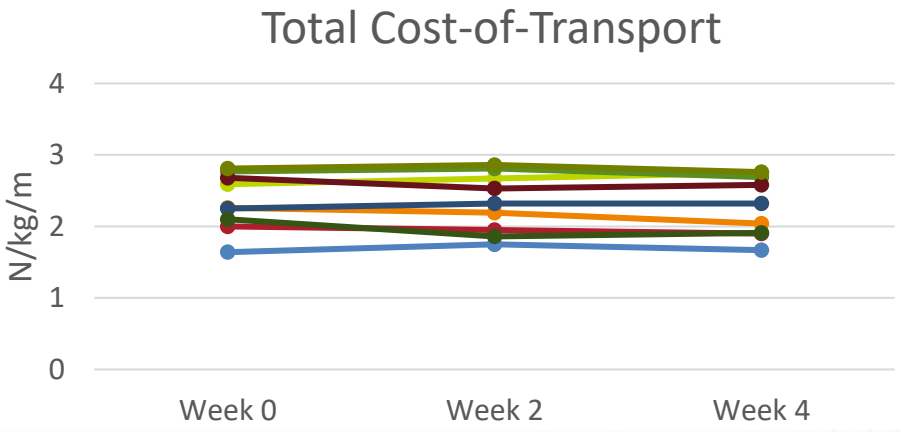

Self-Selected Walking Speed
m/s
1.5
1
0.5
0
Week 0
Week 2
Week 4


Peak Ankle Dorsiflexion
deg
30
25
20
15
10
5
0
Week 0
Week 2
Week 4


Peak Plantar Flexor Moment
N*m/kg
0
-0.5
-1
-1.5
Week 0
Week 2
Week 4


Total Cost-of-Transport
N/kg/m
4
3
2
1
0
Week 0
Week 2
Week 4

Limitations

- Number of participants
- Looking only at peak values may not tell the whole story
- All treadmill data collected at Baseline Walking Speed in order to compare COT across time points

Next Steps

- Analyze outcomes at PD-AFO walking speeds
- Investigate training paradigms

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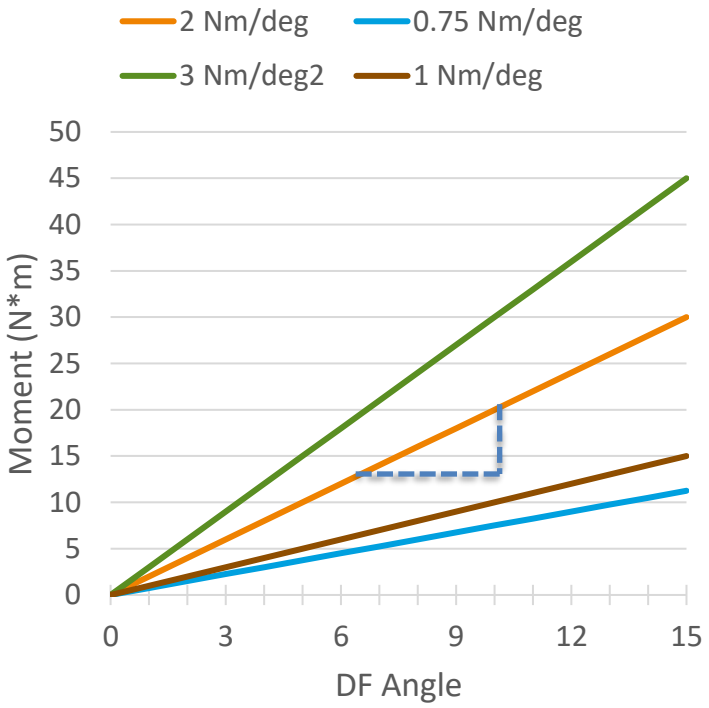
PD-AFOs

- Plantar flexors act like rubber bands to control ankle position during stance (kinematic) and to store and return energy (kinetic)
- PD-AFO stiffness supplements weakened plantar flexors

- Stiffness = resistance to bending
  - Moment increases with dorsiflexion like a torsion spring
  - Varies between individuals



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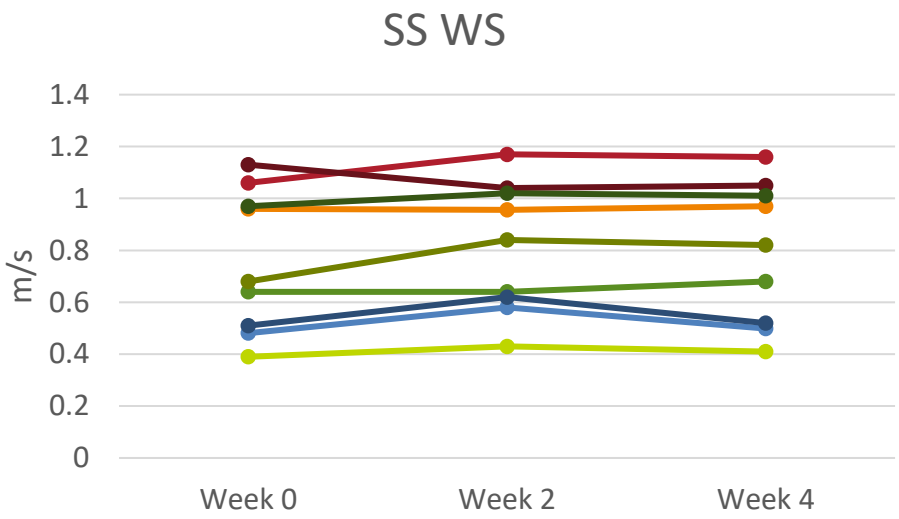
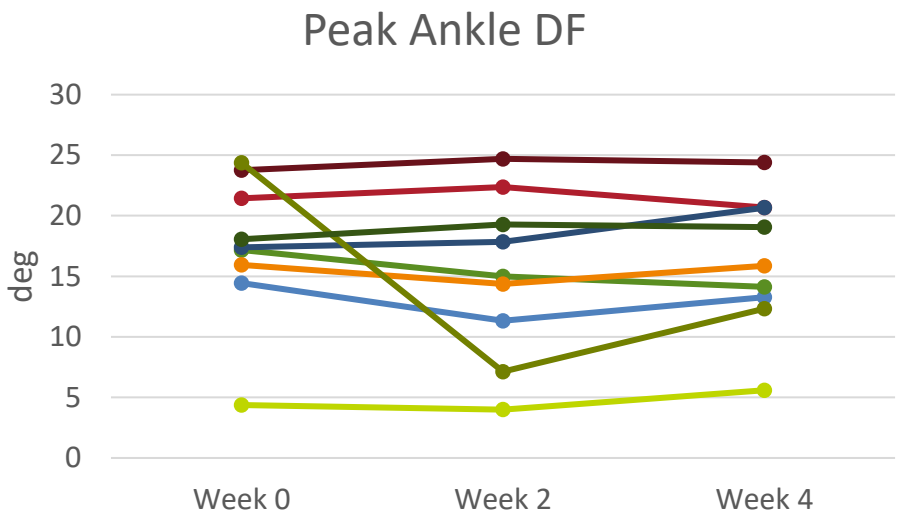
Results – SS WS



Subject	Week 0	Week 2	Week 4
P1	0.481	0.58	0.499
P2	1.06	1.17	1.16
P3	0.39	0.43	0.41
P4	0.64	0.64	0.68
P5	0.96	0.956	0.97
P6	0.51	0.62	0.52
P7	1.13	1.04	1.05
P8	0.68	0.84	0.82
P9	0.97	1.02	1.01
Mean (SD)	**0.76 (0.28)**	**0.81 (0.25)**	**0.79 (0.27)**

Results – Peak Ankle DF



Subject	Week 0	Week 2	Week 4
P1	14.44	11.33	13.27
P2	21.43	22.37	20.68
P3	4.37	3.99	5.59
P4	17.17	14.98	14.13
P5	15.94	14.37	15.87
P6	17.39	17.85	20.66
P7	23.76	24.7	24.41
P8	24.37	7.13	12.32
P9	18.06	19.28	19.07
Mean (SD)	**17.44 (5.98)**	**15.11 (6.82)**	**16.22 (5.66)**

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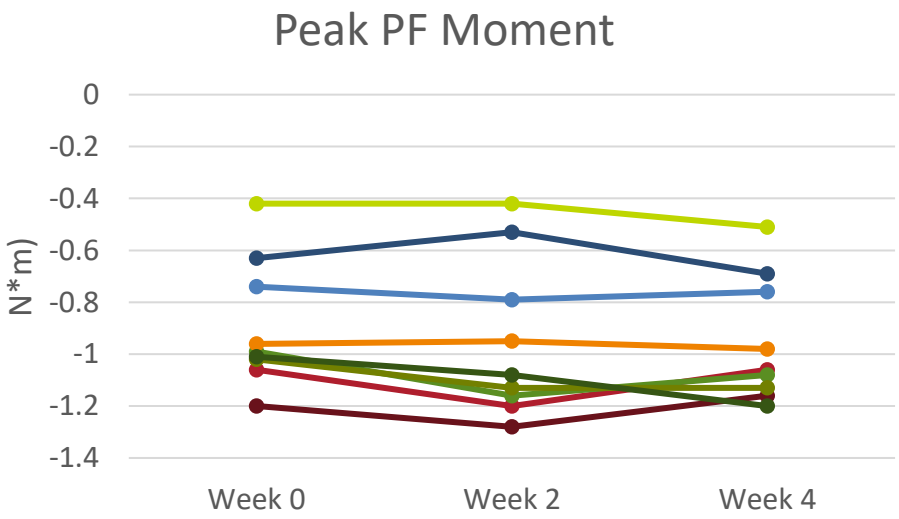
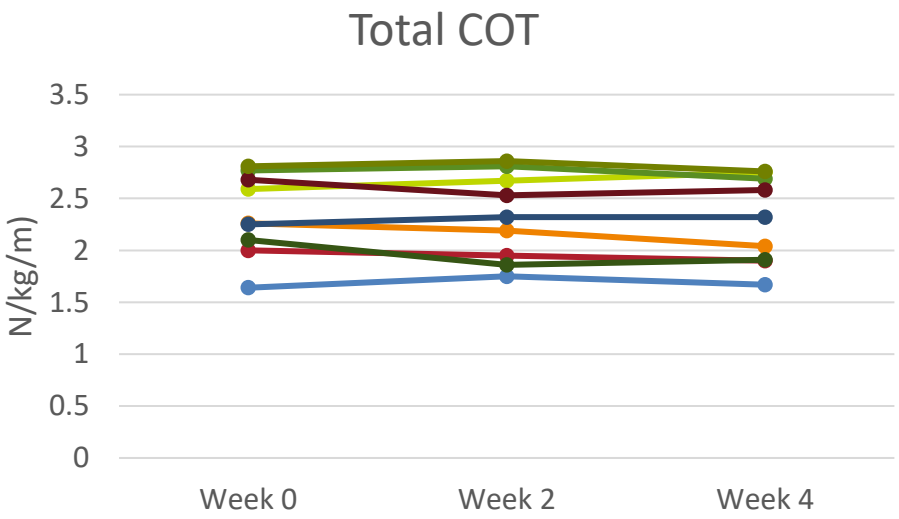
Results – Peak PF Moment



Subject	Week 0	Week 2	Week 4
P1	-0.74	-0.79	-0.76
P2	-1.06	-1.2	-1.06
P3	-0.42	-0.42	-0.51
P4	-0.99	-1.16	-1.08
P5	-0.96	-0.95	-0.98
P6	-0.63	-0.53	-0.69
P7	-1.2	-1.28	-1.16
P8	-1.02	-1.13	-1.13
P9	-1.01	-1.08	-1.2
Mean (SD)	**-0.89 (0.25)**	**-0.95 (0.31)**	**-0.95 (0.24)**

Results – Total COT



Subject	Week 0	Week 2	Week 4
P1	1.64	1.75	1.67
P2	2	1.95	1.9
P3	2.59	2.67	2.75
P4	2.77	2.81	2.69
P5	2.26	2.19	2.04
P6	2.25	2.32	2.32
P7	2.68	2.53	2.58
P8	2.81	2.86	2.76
P9	2.1	1.86	1.91
Mean (SD)	**2.34 (0.40)**	**2.32 (0.42)**	**2.29 (0.42)**

Results

- No meaningful and consistent improvements in the outcome measures over 1 month

Subject	SS WS (m/s)			Peak Ankle DF (deg)			Peak PF Moment (N*m)			Total COT (J/kg/m)		
P1	0.481	0.58	0.499	14.44	11.33	13.27	-0.74	-0.79	-0.76	1.64	1.75	1.67
P2	1.06	1.17	1.16	21.43	22.37	20.68	-1.06	-1.2	-1.06	2	1.95	1.9
P3	0.39	0.43	0.41	4.37	3.99	5.59	-0.42	-0.42	-0.51	2.59	2.67	2.75
P4	0.64	0.64	0.68	17.17	14.98	14.13	-0.99	-1.16	-1.08	2.77	2.81	2.69
P5	0.96	0.956	0.97	15.94	14.37	15.87	-0.96	-0.95	-0.98	2.26	2.19	2.04
P6	0.51	0.62	0.52	17.39	17.85	20.66	-0.63	-0.53	-0.69	2.25	2.32	2.32
P7	1.13	1.04	1.05	23.76	24.7	24.41	-1.2	-1.28	-1.16	2.68	2.53	2.58
P8	0.68	0.84	0.82	24.37	7.13	12.32	-1.02	-1.13	-1.13	2.81	2.86	2.76
P9	0.97	1.02	1.01	18.06	19.28	19.07	-1.01	-1.08	-1.2	2.1	1.86	1.91
Mean	**0.76**	**0.81**	**0.79**	**17.44**	**15.11**	**16.22**	**-0.89**	**-0.95**	**-0.95**	**2.34**	**2.33**	**2.29**
SD	**0.28**	**0.25**	**0.27**	**5.98**	**6.82**	**5.66**	**0.25**	**0.31**	**0.24**	**0.40**	**0.42**	**0.42**